01 / THE PROBLEM

# Bitcoin nodes need a local bouncer.

Raw transaction streams can get noisy. Node operators should be able to express intent locally without rewriting mempool policy or touching consensus.

- Local-only control point
- Prompt-shaped policy
- Suspicious traffic gets checked

Dashboard signal

**candidate rawtx enters**

| | |
|---|---|
| **Wallet** | rawtx |
| **Bouncer** | judge |
| **Gate Node** | only if allowed |

02 / THE IDEA

# A tiny agent decides: pass, hold, drop, shadow drop.

Bitcoin Bouncer summarizes a transaction, checks it against Bitcoin Core, then asks a local model to choose from a few explicit actions.

Allowed verbs

PASS
HOLD
DROP
SHADOW

Featured action

shadow_drop(txid)

**Small tool surface**

**Fast local decisions**

**Truthful audit log**

03 / THE SHADOW REALM

# Return a txid. Broadcast nothing. Keep the record.

Shadow drop returns a normal-looking response while withholding the transaction from the Gate Node and storing it in the Shadow Realm for review.

**Submitter gets a txid**

**Gate Node sees nothing**

**Shadow Realm keeps context**

Dashboard outcome

**Shadow dropped**

Demo line

**stored in the Shadow Realm**

04 / DEMO FLOW

# Run smoke, force shadow drop, then check witnesses.

The demo shows the difference between a returned txid and real propagation. Witness nodes make that visible.

| | |
|---|---|
| 1 | Start runtime |
| 2 | Force shadow_drop |
| 3 | Show "not visible" |

Best live moment

**txid response -> no propagation**

**Smoke: happy path**

**Forced actions: shadow drop**

**Witnesses: absent by design**

05 / THE PUNCHLINE

# A programmable membrane for Bitcoin traffic.

The harness stays simple. The model gets just enough agency to protect the local path and explain exactly what happened.

**Submission gate today**

**Shadow Realm receipts**

**Built for local regtest demos**

One sentence

**Bitcoin Bouncer lets your node decide what gets through.**

Hackathon takeaway

**live**
**local**
**auditable**